

ICT702 Data Visualisation

Selecting a Chart Type

Module 2



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



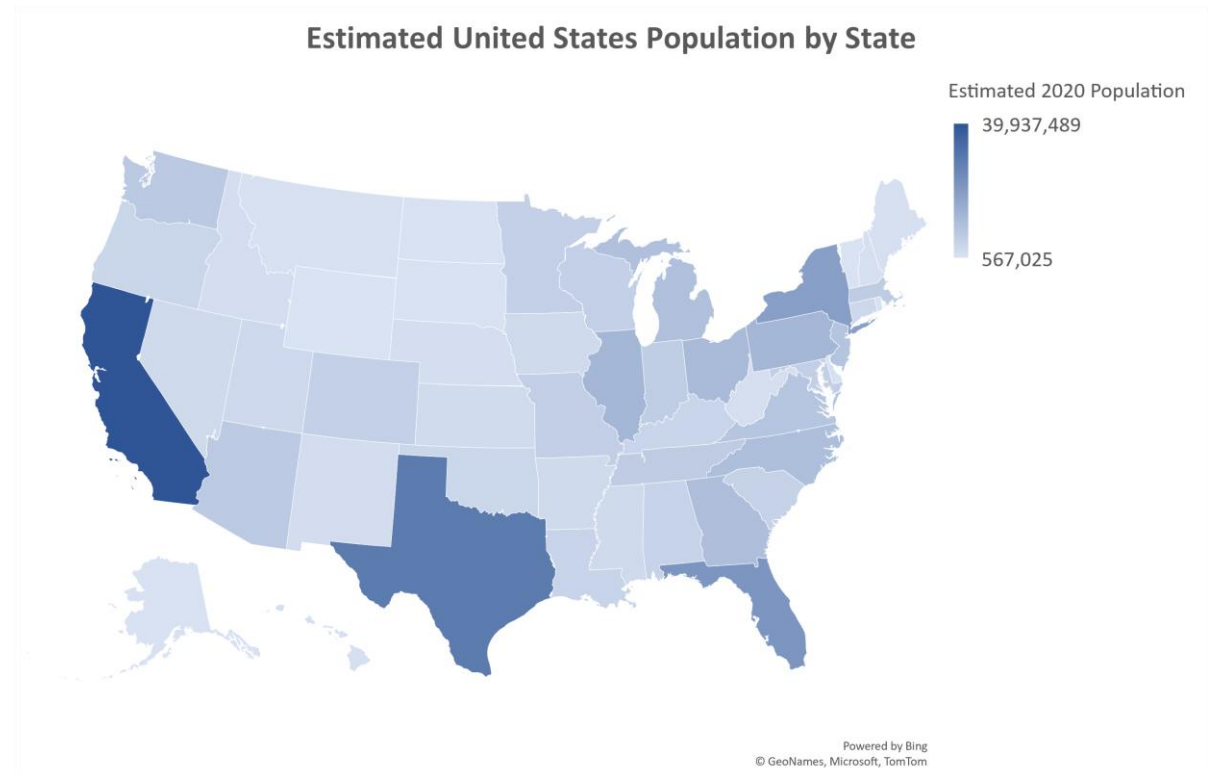
Part 2

Choropleth Maps

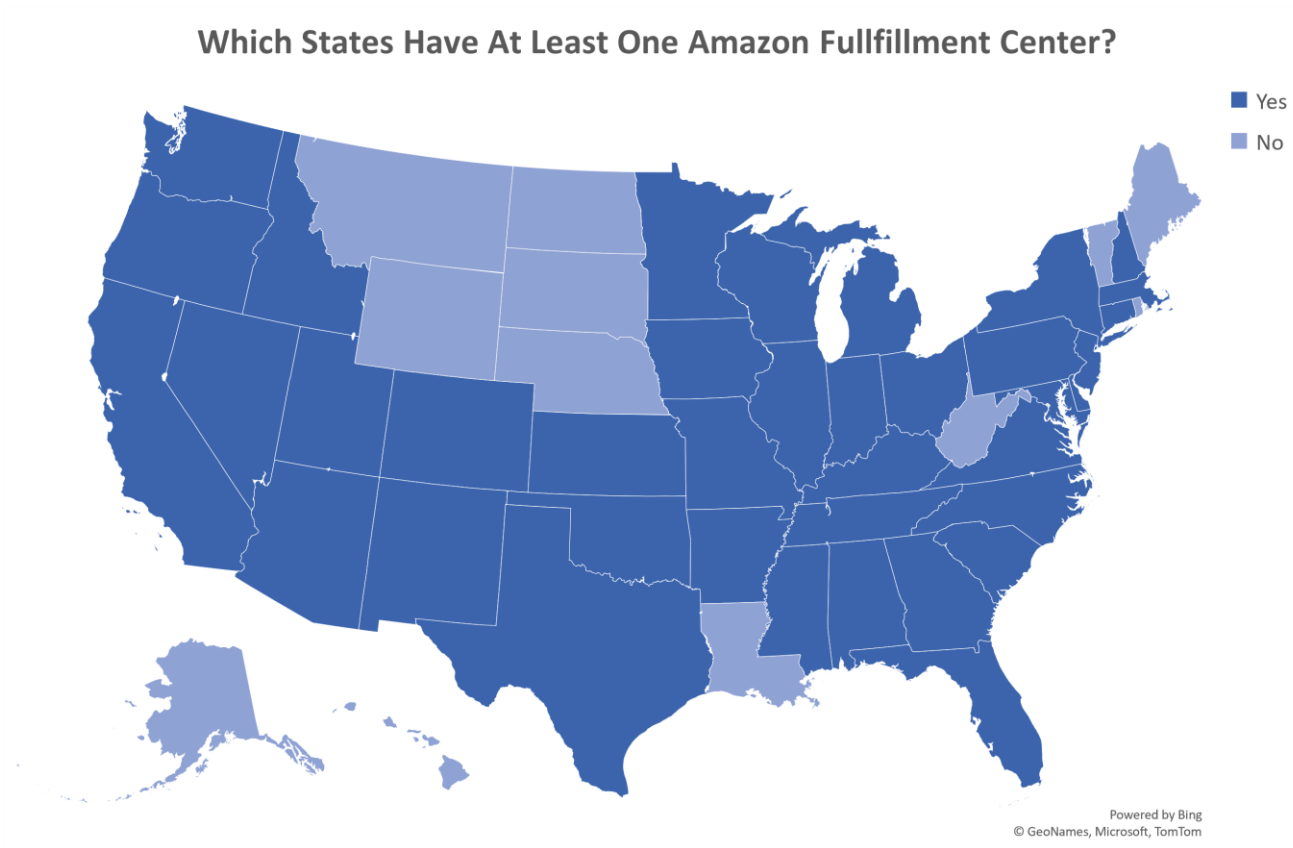
Choropleth maps can be built for quantitative or categorical data.

	A	B
1	State	Estimated 2020 Population
2	California	39,937,489
3	Texas	29,472,295
4	Florida	21,992,985
5	New York	19,440,469
6	Pennsylvania	12,820,878
7	Illinois	12,659,682
8	Ohio	11,747,694
9	Georgia	10,736,059
10	North Carolina	10,611,862

The map indicates the states with the highest population.



Choropleth Maps for Categorical Data



- The map shows that Amazon has at least one fulfillment center in 38 states.
- States without a fulfillment center tend to be either relatively sparsely populated or a geographic outlier.
- Amazon has many fulfillment centers to ensure quick customer delivery times for many of the products it sells.

Heat Maps

A heat map is a 2D graphical representation of a data table that uses shades of colour to indicate magnitude.

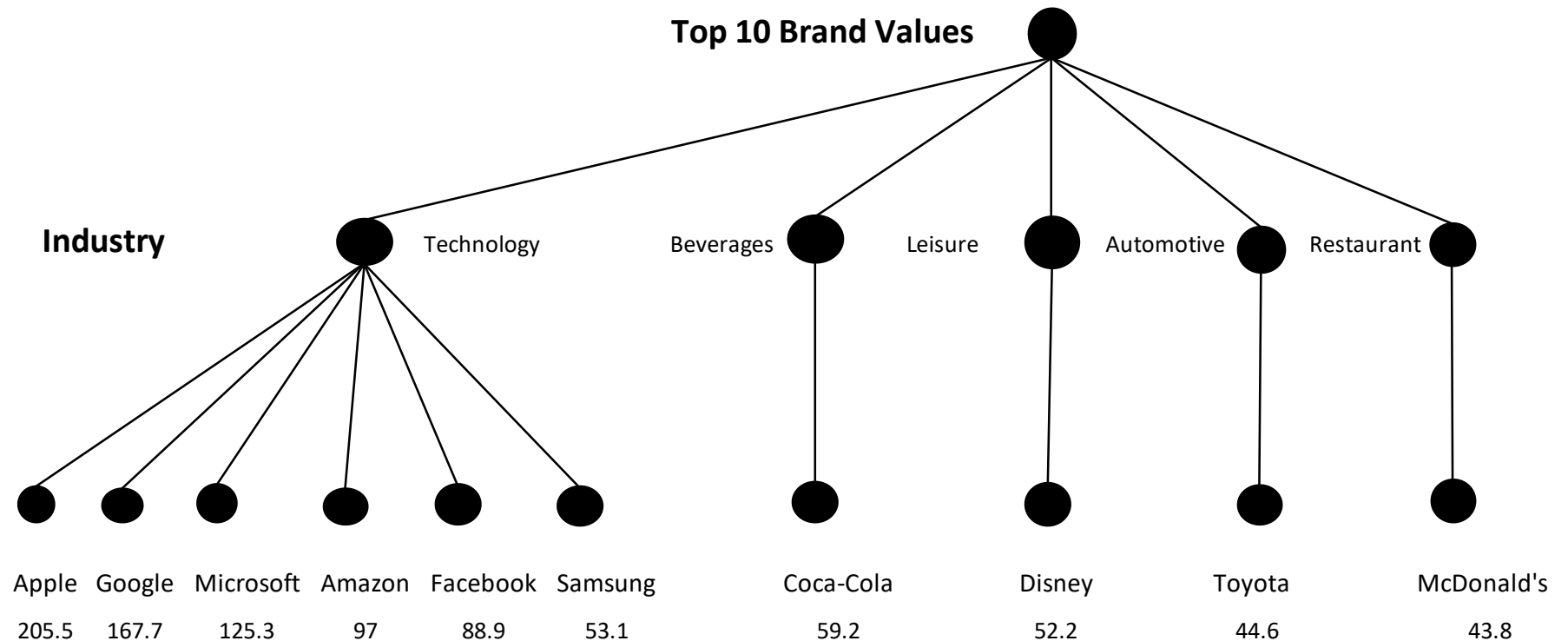
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2	St. Louis	-2%	-1%	-1%	0%	2%	4%	3%	5%	6%	7%	8%	8%
3	Phoenix	5%	4%	4%	2%	2%	-2%	-5%	-8%	-6%	-5%	-7%	-8%
4	Albany	-5%	-6%	-4%	-5%	-2%	-5%	-5%	-3%	-1%	-2%	-1%	-2%
5	Austin	16%	15%	15%	16%	18%	17%	14%	15%	16%	19%	18%	16%
6	Cincinnati	-9%	-6%	-7%	-3%	3%	6%	8%	11%	10%	11%	13%	11%
7	San Francisco	2%	4%	5%	8%	4%	2%	4%	3%	1%	-1%	1%	2%
8	Seattle	7%	7%	8%	7%	5%	4%	2%	0%	-2%	-4%	-6%	-5%
9	Chicago	5%	3%	2%	6%	8%	7%	8%	5%	8%	10%	9%	8%
10	Atlanta	12%	14%	13%	17%	12%	11%	8%	7%	7%	8%	5%	3%
11	Miami	2%	3%	0%	1%	-1%	-4%	-6%	-8%	-11%	-13%	-11%	-10%
12	Minneapolis	-6%	-6%	-8%	-5%	-6%	-5%	-5%	-7%	-5%	-2%	-1%	-2%
13	Denver	5%	4%	1%	1%	2%	3%	1%	-1%	0%	1%	2%	3%
14	Salt Lake City	7%	7%	7%	13%	12%	8%	5%	9%	10%	9%	7%	6%
15	Raleigh	4%	2%	0%	5%	4%	3%	5%	5%	9%	11%	8%	6%
16	Boston	-5%	-5%	-3%	4%	-5%	-4%	-3%	-1%	1%	2%	3%	5%
17	Pittsburgh	-6%	-6%	-4%	-5%	-3%	-3%	-1%	-2%	-2%	-1%	-2%	-1%

Hierarchy of a Treemap

The Hierarchical Tree Structure of the Top Ten Brand Values Data

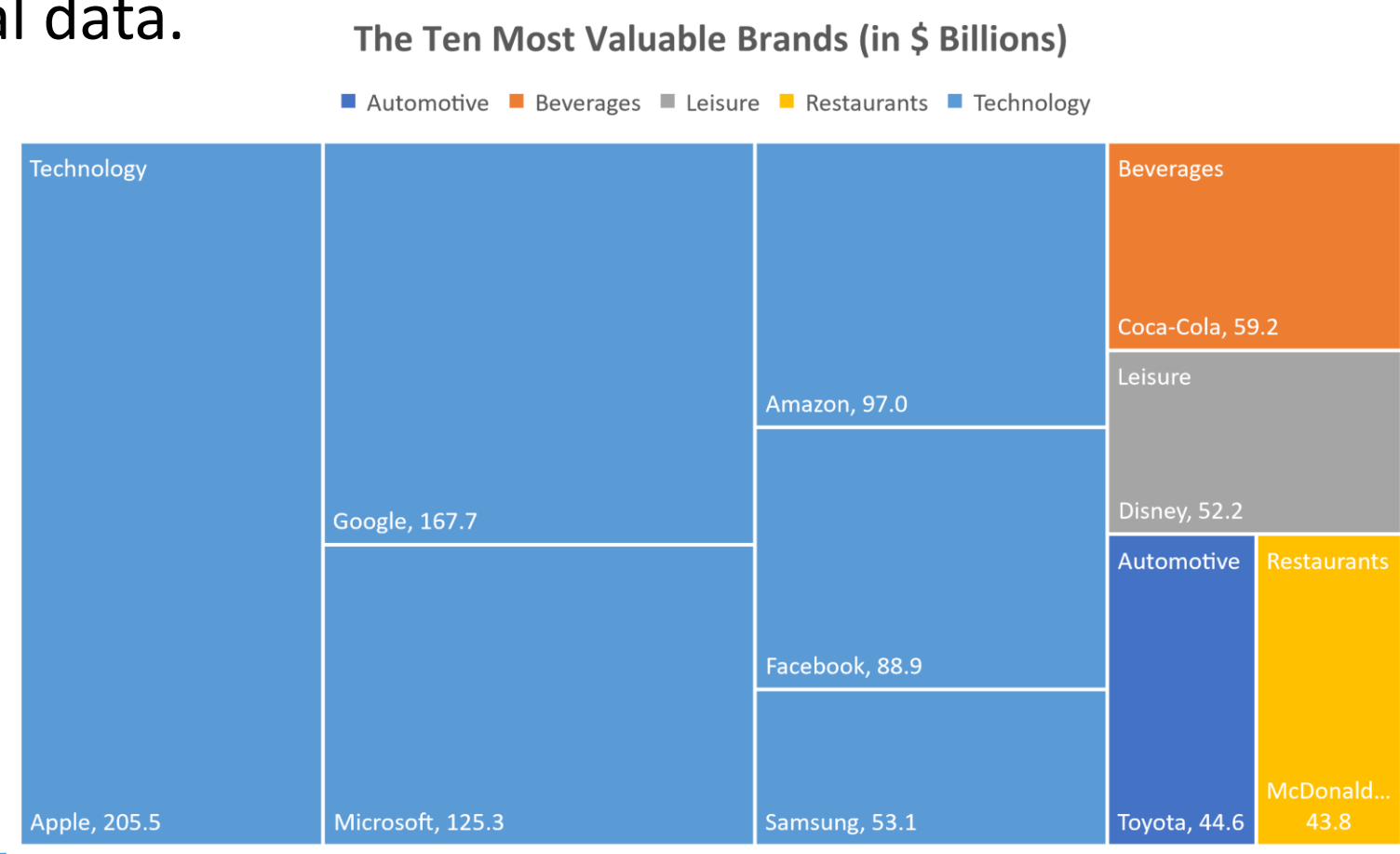
	A	B	C
1	Industry	Brand	Value (in \$ Billions)
2	Technology	Apple	205.5
3	Technology	Google	167.7
4	Technology	Microsoft	125.3
5	Technology	Amazon	97.0
6	Technology	Facebook	88.9
7	Beverages	Coca-Cola	59.2
8	Technology	Samsung	53.1
9	Leisure	Disney	52.2
10	Automotive	Toyota	44.6
11	Restaurants	McDonald's	43.8

Company
Value (in \$ Billions)



Editing a Treemap

A treemap is a chart that uses the size, colour, and arrangement of rectangles to display the magnitudes of a quantitative variable for hierarchical data.



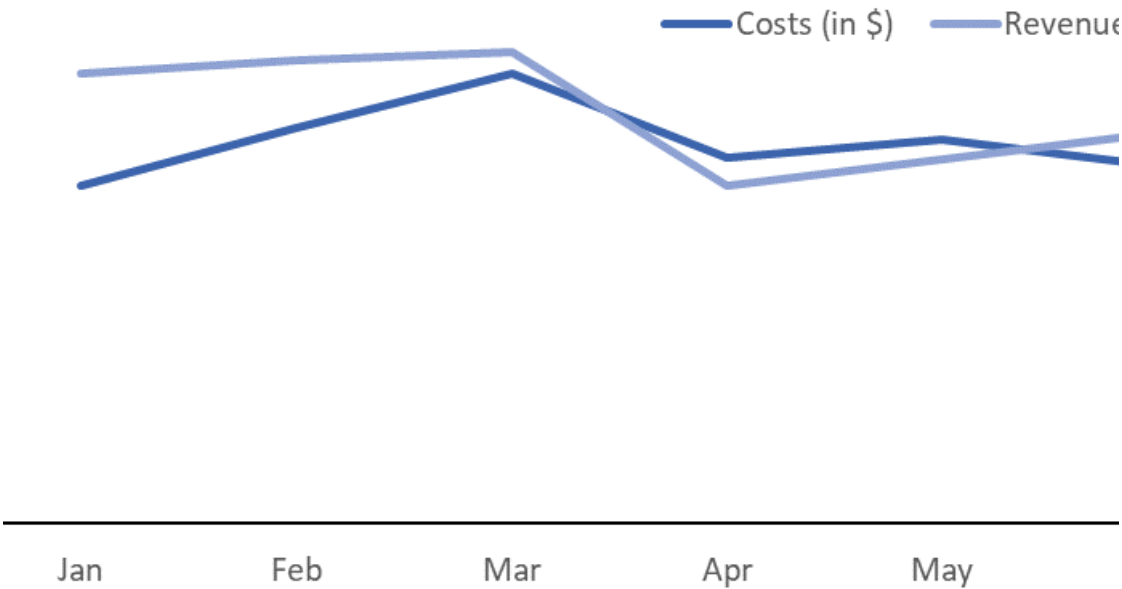
Tables versus Charts

Table Showing Exact Values for Costs and Revenues by Month for Gossamer Industries

	Month			
Jan	Feb	Mar	Apr	May
3,123	56,458	64,125	52,158	54,712
4,124	66,125	67,125	48,178	51,789

Line Chart of Monthly Costs and Revenues

Gossamer Industries: Revenue and Costs

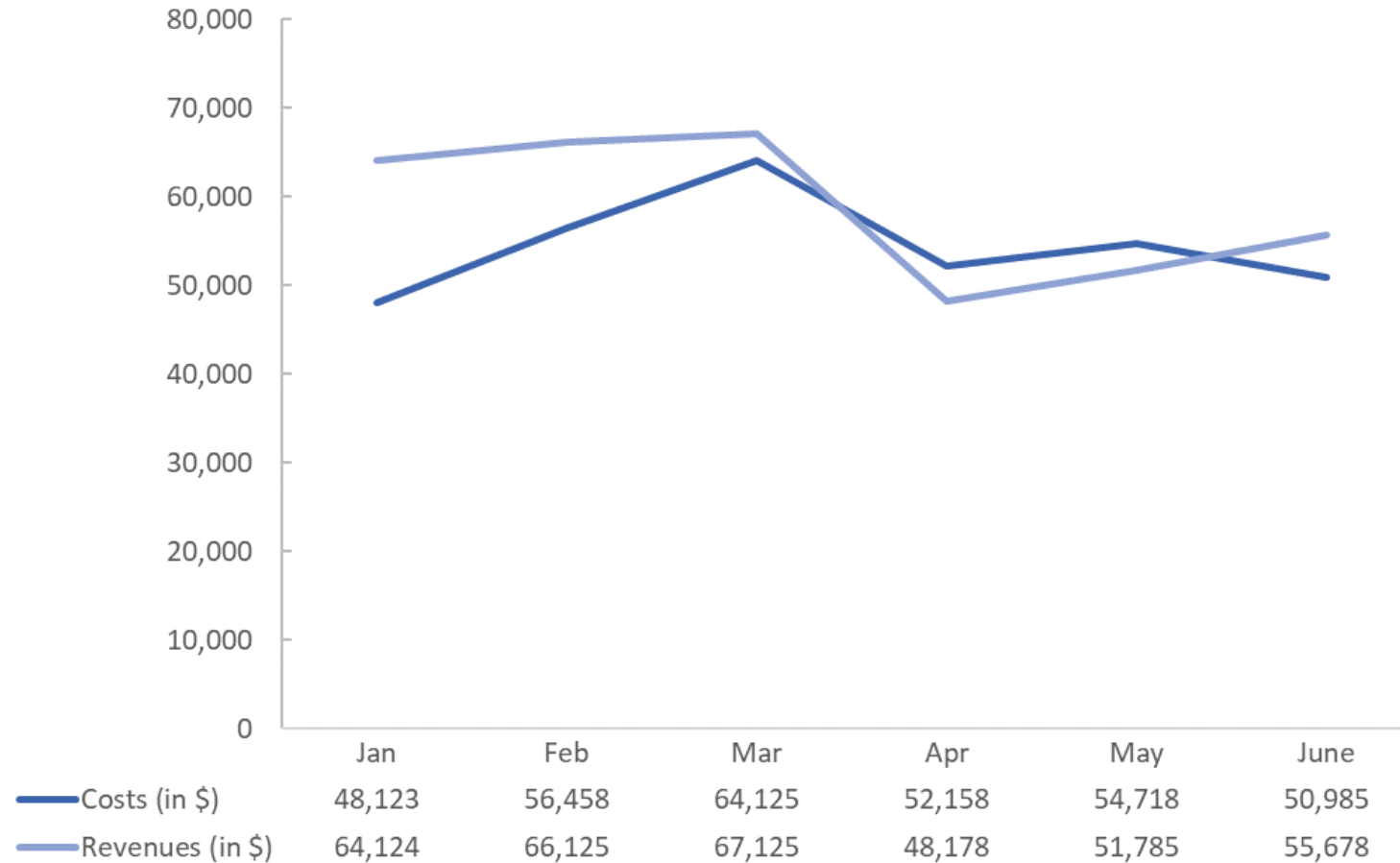


Tables should be used when:

- specific numerical values are required
- precise comparisons between different values need to be made
- values being displayed have different units or very different magnitudes

Combined Line Charts and Tables

Gossamer Industries: Revenue and Costs

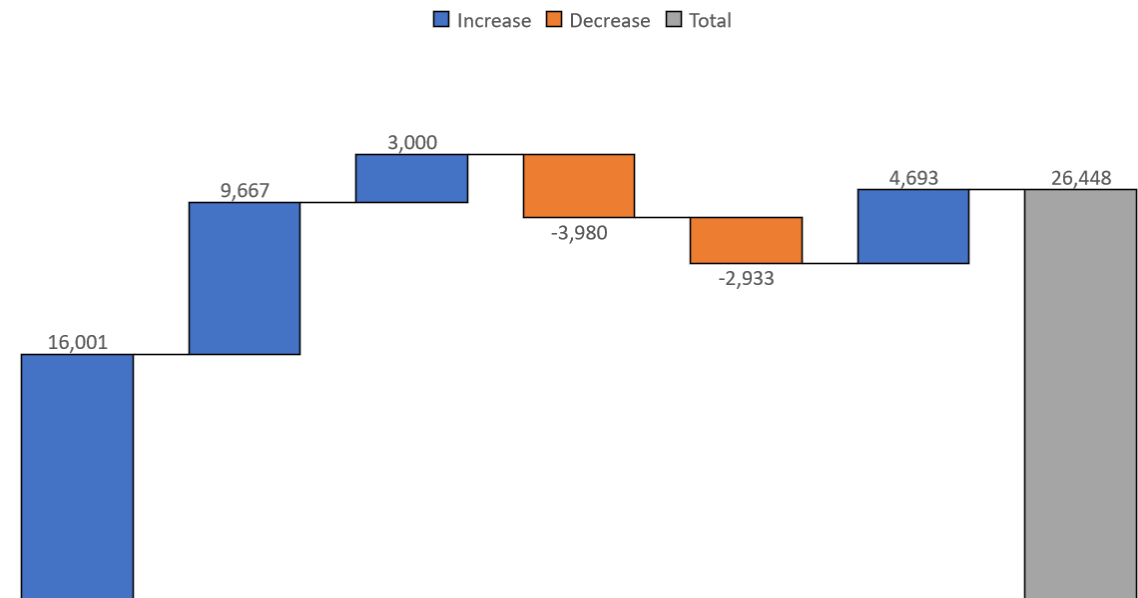


Waterfall Charts

GossamerGP data file for Gossamer costs, revenues, and gross profit

1				Month				
2		Jan	Feb	Mar	Apr	May	June	Total
3	Costs (in \$)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
4	Revenues (in \$)	64,124	66,125	67,125	48,178	51,785	55,678	353,015
5	Gross Profit (in \$)	16,001	9,667	3,000	-3,980	-2,933	4,693	26,448

A waterfall chart for the gossamer gross profit data

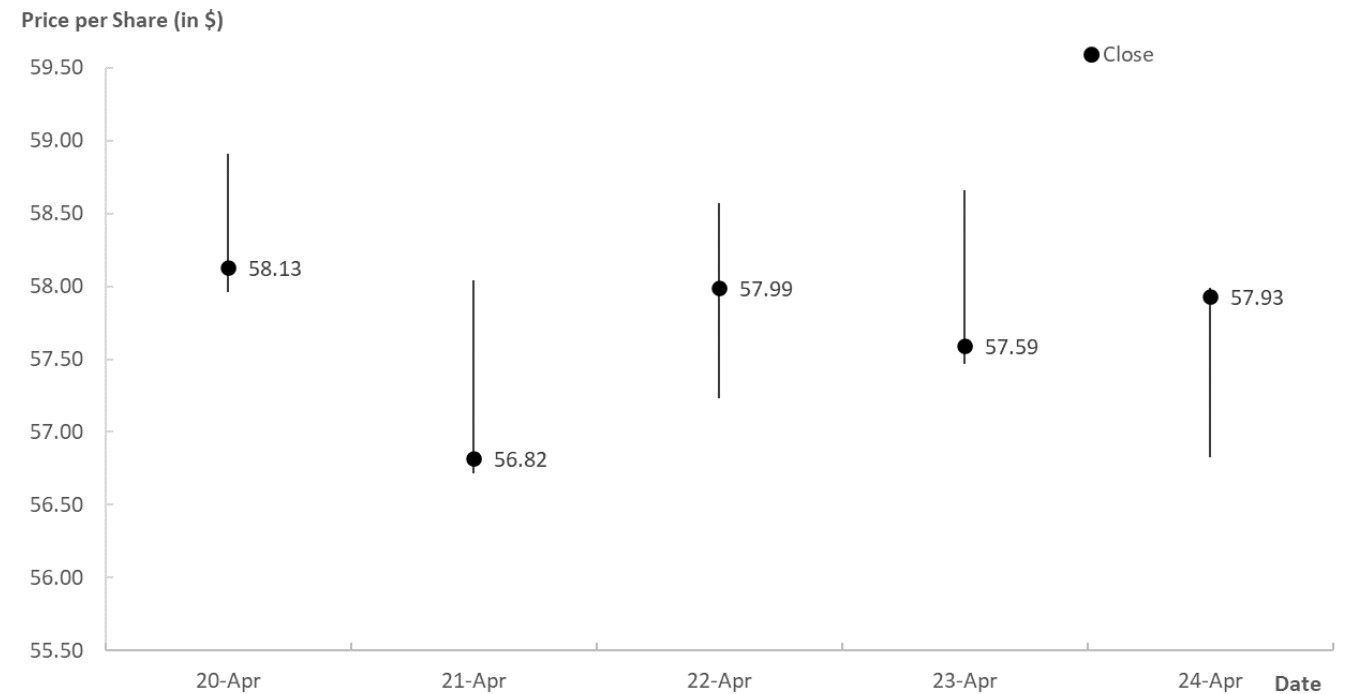


Stock Charts

Verizon stock price data file

	A	B	C	D	E
1	Date	Open	High	Low	Close
2	20-Apr	58.10	58.91	57.96	58.13
3	21-Apr	57.39	58.04	56.72	56.82
4	22-Apr	57.41	58.57	57.23	57.99
5	23-Apr	58.12	58.66	57.47	57.59
6	24-Apr	57.64	57.99	56.83	57.93

A high-low-close stock chart for Verizon Wireless

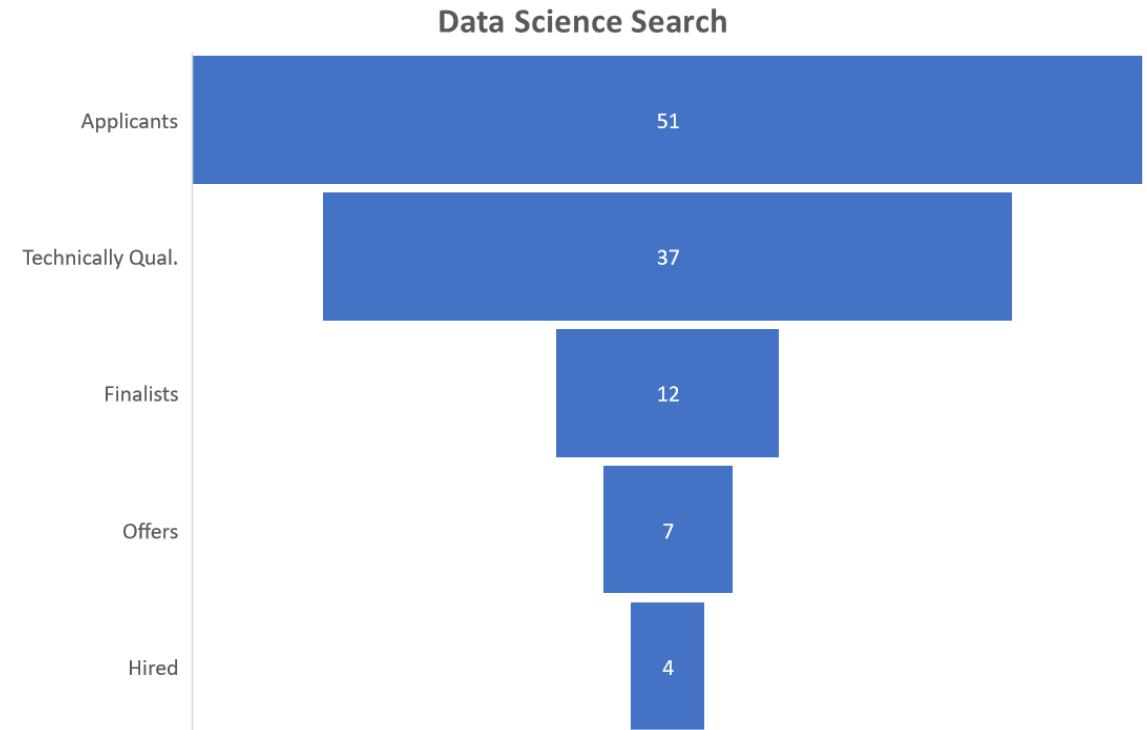


Funnel Charts

DataScienceSearch data for hiring

	A	B
1	Stage	Number of Applicants
2	Applicants	51
3	Technically Qualified	37
4	Finalists	12
5	Offers	7
6	Hired	4

A funnel chart for Data Scientist hiring



End of Part 2

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